Clarence AUERBACH

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Master 1 — Mathematics & Artificial Intelligence

Candidate for a Research Internship

Availability: March 23 – August 31, 2027 (4 to 5 months) · Focus: Statistical Learning

Education

- **Master's in Mathematics and Artificial Intelligence** 2026–2027 (1 year)
  Université Paris-Saclay
  – **Relevant Coursework:** Advanced Statistics, Machine Learning Project, Unsupervised and Supervised Learning, Deep Learning Project.

- **Double Bachelor's Degree in Mathematics and Computer Science** (With Honors) 2023–2026
  Université Paris Cité
  – **Relevant Coursework:** Statistics, Optimization with Constraints, Algorithmics, Functional and Object-Oriented Programming.

Projects

- **Transformer with PyTorch** September 2026
  Personal project
  – Implementation of a Transformer architecture in PyTorch (multi-head attention, positional encoding, training loop), trained and evaluated on a sequence task.

- **Linearly-Solvable MDPs** May 2026
  Undergraduate Research Project (Supervisor: Simon Coste)
  – Study of Todorov's linearly-solvable MDPs: Python implementation of Z-learning, Q-learning and Z-iteration.

- **FNN from Scratch** November 2025
  Personal project
  – Feedforward neural network implemented from scratch, without deep learning libraries: forward pass, backpropagation and gradient-descent training.

- **3D Viewer with OpenGL** July 2025
  Personal project
  – Real-time 3D viewer built with OpenGL: shader-based rendering pipeline, camera controls and navigation.

→ Others on clarenceauerbach.github.io

Technical Skills

Programming & Tools: Python (`PyTorch`, `scikit-learn`, `NumPy`), R, Java, C, Lean, Git, Docker, VS Code, RStudio.

Applied Mathematics: Convex & Non-Convex Optimization, Applied Probability, Stochastic Processes & Markov Chains, Bayesian Statistics, High-Dimensional Linear Algebra, Formal Logic.

Languages & Personal Interests

Languages: French (Native), English (Fluent: C2), Spanish (B2).

Interests: Chess (2000 Elo), Variant Sudoku (solved Ten Ring Circus), Rock Climbing (6b+).